

Practical 2

Aim

To perform the **ETL (Extract, Transform, Load)** process using SQL Server.

Code

```
CREATE DATABASE ETL_Practical;
GO
USE ETL_Practical;
GO
CREATE TABLE Student_Data
(
    Student_ID INT,
    Student_Name VARCHAR(50),
    Age INT,
    City VARCHAR(50),
    Marks DECIMAL(5,2)
);
GO
INSERT INTO Student_Data
VALUES
(1, 'Rahul', 20, 'Mumbai', 85),
(2, 'Sneha', 21, 'mumbai', 92),
(3, 'Amit', NULL, 'Pune', 78),
(4, 'Priya', 20, NULL, 88),
(5, 'Rohan', 22, 'PUNE', 65);
GO
```

• Extract

```
SELECT
    Student_ID,
    Student_Name,
    Age,
    City,
    Marks
FROM Student_Data;
GO
```

• Transform

```
SELECT
    Student_ID,
    Student_Name,
    ISNULL(Age, 0) AS Age,

    CASE
        WHEN UPPER(City) = 'MUMBAI' THEN 'Mumbai'
        WHEN UPPER(City) = 'PUNE' THEN 'Pune'
        ELSE ISNULL(City, 'Unknown')
```

```
END AS City,

Marks,

CASE
  WHEN Marks >= 85 THEN 'Excellent'
  WHEN Marks >= 70 THEN 'Good'
  ELSE 'Needs Improvement'
END AS Performance_Category

FROM Student_Data;
GO
```

- **Load**

```
CREATE TABLE Student_Final
(
  Student_ID INT,
  Student_Name VARCHAR(50),
  Age INT,
  City VARCHAR(50),
  Marks DECIMAL(5,2),
  Performance_Category VARCHAR(30)
);
GO

INSERT INTO Student_Final
(
  Student_ID,
  Student_Name,
  Age,
  City,
  Marks,
  Performance_Category
)
SELECT
  Student_ID,
  Student_Name,
  ISNULL(Age, 0),

  CASE
    WHEN UPPER(City) = 'MUMBAI' THEN 'Mumbai'
    WHEN UPPER(City) = 'PUNE' THEN 'Pune'
    ELSE ISNULL(City, 'Unknown')
  END,

  Marks,

  CASE
    WHEN Marks >= 85 THEN 'Excellent'
    WHEN Marks >= 70 THEN 'Good'
    ELSE 'Needs Improvement'
```

```

END

FROM Student_Data;
GO

SELECT * FROM Student_Final;
GO

```

Output

Extract

	Student_ID	Student_Name	Age	City	Marks
1	1	Rahul	20	Mumbai	85.00
2	2	Sneha	21	mumbai	92.00
3	3	Amit	NULL	Pune	78.00
4	4	Priya	20	NULL	88.00
5	5	Rohan	22	PUNE	65.00

Transform

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 Results

 Messages

	Student_ID	Student_Name	Age	City	Marks	Performance_Category
1	1	Rahul	20	Mumbai	85.00	Excellent
2	2	Sneha	21	Mumbai	92.00	Excellent
3	3	Amit	0	Pune	78.00	Good
4	4	Priya	20	Unknown	88.00	Excellent
5	5	Rohan	22	Pune	65.00	Needs Improvement

Load

	Student_ID	Student_Name	Age	City	Marks	Performance_Category
1	1	Rahul	20	Mumbai	85.00	Excellent
2	2	Sneha	21	Mumbai	92.00	Excellent
3	3	Amit	0	Pune	78.00	Good
4	4	Priya	20	Unknown	88.00	Excellent
5	5	Rohan	22	Pune	65.00	Needs Improvement